[BUG] Incomplete Operator Name Mapping for Controlled RX Gate

https://github.com/PennyLaneAI/pennylane/issues/4660

ROW 76

[BUG] Incomplete Operator Name Mapping for Controlled RX Gate #4660

New issue

Closed #4697

c-3543 opened on Oct 7, 2023

Expected behavior

Great question. The way operators get translated from PL to Qiskit is via a dictionary lookup given the operator's name in PL (see here for the source code). The controlled RX gate has a different name in PennyLane depending on how you define it:

> > > op = qml.CRX(0.1, wires=[0, 1])
> > > op.name
> > > 'CRX'
> > > op = qml.ctrl(qml.RX(0.1, 0), control=[1])
> > > op.name
> > > 'C(RX)'

Actual behavior

There is a mapping for when the name is CRX, but not for C(RX)

Additional information

Info here:
https://discuss.pennylane.ai/t/pennylane-ibmq-conversion-error/3498

Source code

```
import pennylane as qml
from pennylane import numpy as np

property_prices = [4, 8, 6, 3, 12, 15] # total 48
variables_wires = [0, 1, 2, 3, 4, 5]

aux_oracle_wires = [6, 7, 8 ,9 ,10, 11]

def oracle(variables_wires, aux_oracle_wires):

    def add_k_fourier(k, wires):
        for j in range(len(wires)):
            qml.RZ(k * np.pi / (2**j), wires=wires[j])

    def value_second_sibling():

        qml.QFT(wires = aux_oracle_wires)

        for wire in variables_wires:
            qml.ctrl(add_k_fourier, control = wire)(property_prices[wire], wires = aux_oracle_wires)

        qml.adjoint(qml.QFT)(wires = aux_oracle_wires)

    value_second_sibling()
    qml.FlipSign(sum(property_prices) // 2, wires = aux_oracle_wires)
    qml.adjoint(value_second_sibling)()

dev = qml.device('qiskit.ibmq', wires=12, backend='ibmq_qasm_simulator')

@qml.qnode(dev)
def circuit():

    # step 1
    for wire in variables_wires:
        qml.Hadamard(wires = wire)

    # step 2
    oracle(variables_wires, aux_oracle_wires)

    # step 3
    qml.GroverOperator(wires = variables_wires)

    return qml.probs(wires = variables_wires)


import matplotlib.pyplot as plt
```

Assignees
No one assigned

Labels
bug 🐛

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
Decompositions for multi-controlled single qubit special unitary ops
PennyLaneAI/pennylane

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Participants

Visual S

```
values = circuit()
plt.bar(range(len(values)), values)
```

Tracebacks

```
# Put full error message here

---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[5], line 3
      1 import matplotlib.pyplot as plt
----> 3 values = circuit()
      4 plt.bar(range(len(values)), values)

File ~\anaconda3\Lib\site-packages\pennylane\qnode.py:989, in QNode.__call__(self, *args, **kwargs)
    986     self.execute_kwargs.pop("mode")
    988 # pylint: disable=unexpected-keyword-arg
--> 989 res = qml.execute(
    990     (self._tape,),
    991     device=self.device,
    992     gradient_fn=self.gradient_fn,
    993     interface=self.interface,
    994     transform_program=self.transform_program,
    995     gradient_kwargs=self.gradient_kwargs,
    996     override_shots=override_shots,
    997     **self.execute_kwargs,
    998 )
   1000 res = res[0]
   1002 # convert result to the interface in case the qfunc has no parameters

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:757, in execute(tapes, device, gradient_f
    754     raise ValueError("Gradient transforms cannot be used with grad_on_execution=True")
    756 ml_boundary_execute = _get_ml_boundary_execute(interface, _grad_on_execution)
--> 757 results = ml_boundary_execute(
    758     tapes, device, execute_fn, gradient_fn, gradient_kwargs, _n=1, max_diff=max_diff
    759 )
    761 results = batch_fn(results)
    762 return program_post_processing(results)

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\autograd.py:317, in execute(tapes, device, execute_fn,
    312 # pylint misidentifies autograd.builtins as a dict
    313 # pylint: disable=no-member
    314 parameters = autograd.builtins.tuple(
    315     [autograd.builtins.list(t.get_parameters()) for t in tapes]
    316 )
--> 317 return _execute(
    318     parameters,
    319     tapes=tapes,
    320     device=device,
    321     execute_fn=execute_fn,
    322     gradient_fn=gradient_fn,
    323     gradient_kwargs=gradient_kwargs,
    324     _n=_n,
    325     max_diff=max_diff,
    326 )[0]

File ~\anaconda3\Lib\site-packages\autograd\tracer.py:48, in primitive.<locals>.f_wrapped(*args, **kwargs)
     46     return new_box(ans, trace, node)
     47 else:
---> 48     return f_raw(*args, **kwargs)

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\autograd.py:378, in _execute(parameters, tapes, device
    360 if logger.isEnabledFor(logging.DEBUG):
    361     logger.debug(
    362         "Entry with args=(parameters=%s, tapes=%s, device=%s, execute_fn=%s, gradient_fn=%s, gradient_
    363         parameters,
   (...)
    375         "::L".join(str(i) for i in inspect.getouterframes(inspect.currentframe(), 2)[1][1:3]),
    376     )
--> 378 res, jacs = execute_fn(tapes, **gradient_kwargs)
    380 return res, jacs

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:623, in execute.<locals>.inner_execute_wi
    622 def inner_execute_with_empty_jac(tapes, **_):
--> 623     return (inner_execute(tapes), [])

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:255, in _make_inner_execute.<locals>.inne
    253 if numpy_only:
    254     tapes = tuple(qml.transforms.convert_to_numpy_parameters(t) for t in tapes)
--> 255 return cached_device_execution(tapes)
```

```
File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:377, in cache_execute.<locals>.wrapper(ta
    372         return (res, []) if return_tuple else res
    374 else:
    375     # execute all unique tapes that do not exist in the cache
    376     # convert to list as new device interface returns a tuple
--> 377     res = list(fn(tuple(execution_tapes.values()), **kwargs))
    379 final_res = []
    381 for i, tape in enumerate(tapes):

File ~\anaconda3\Lib\contextlib.py:81, in ContextDecorator.__call__.<locals>.inner(*args, **kwds)
     78 @wraps(func)
     79 def inner(*args, **kwds):
     80     with self._recreate_cm():
---> 81         return func(*args, **kwds)

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\ibmq.py:91, in IBMQDevice.batch_execute(self, circuits)
     90 def batch_execute(self, circuits):  # pragma: no cover, pylint:disable=arguments-differ
---> 91     res = super().batch_execute(circuits, timeout=self.timeout_secs)
     92     if self.tracker.active:
     93         self._track_run()

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:473, in QiskitDevice.batch_execute(self,
    470 def batch_execute(self, circuits, timeout: int = None):
    471     # pylint: disable=missing-function-docstring
--> 473     compiled_circuits = self.compile_circuits(circuits)
    475     # Send the batch of circuit objects using backend.run
    476     self._current_job = self.backend.run(compiled_circuits, shots=self.shots, **self.run_args)

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:462, in QiskitDevice.compile_circuits(sel
    458 for circuit in circuits:
    459     # We need to reset the device here, else it will
    460     # not start the next computation in the zero state
    461     self.reset()
--> 462     self.create_circuit_object(circuit.operations, rotations=circuit.diagonalizing_gates)
    464     compiled_circ = self.compile()
    465     compiled_circ.name = f"circ{len(compiled_circuits)}"

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:273, in QiskitDevice.create_circuit_objec
    261 """Builds the circuit objects based on the operations and measurements
    262 specified to apply.
    263
   (...)
    269         pre-measurement into the eigenbasis of the observables.
    270 """
    271 rotations = kwargs.get("rotations", [])
--> 273 applied_operations = self.apply_operations(operations)
    275 # Rotating the state for measurement in the computational basis
    276 rotation_circuits = self.apply_operations(rotations)

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:322, in QiskitDevice.apply_operations(sel
    318         par[idx] = p.tolist()
    320 operation = operation.name
--> 322 mapped_operation = self._operation_map[operation]
    324 self.qubit_state_vector_check(operation)
    326 qregs = [self._reg[i] for i in device_wires.labels]

KeyError: 'C(RZ)'
```

System information

```
Name: PennyLane
Version: 0.32.0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: GitHub - PennyLaneAI/pennylane: PennyLane is a cross-platform Python library for differentiable prc
Author:
Author-email:
License: Apache License 2.0
Location: C:\Users\Carter\anaconda3\Lib\site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: PennyLane-Lightning, PennyLane-qiskit

Platform info: Windows-10-10.0.22621-SP0
Python version: 3.11.5
Numpy version: 1.23.5
Scipy version: 1.11.1
Installed devices:

default.gaussian (PennyLane-0.32.0)
default.mixed (PennyLane-0.32.0)
default.qubit (PennyLane-0.32.0)
default.qubit.autograd (PennyLane-0.32.0)
```

Microsoft Pov

```
default.qubit.autograd (PennyLane-0.32.0)
default.qubit.jax (PennyLane-0.32.0)
default.qubit.tf (PennyLane-0.32.0)
default.qubit.torch (PennyLane-0.32.0)
default.qutrit (PennyLane-0.32.0)
null.qubit (PennyLane-0.32.0)
lightning.qubit (PennyLane-Lightning-0.32.0)
qiskit.aer (PennyLane-qiskit-0.32.0)
qiskit.basicaer (PennyLane-qiskit-0.32.0)
qiskit.ibmq (PennyLane-qiskit-0.32.0)
qiskit.ibmq.circuit_runner (PennyLane-qiskit-0.32.0)
qiskit.ibmq.sampler (PennyLane-qiskit-0.32.0)
qiskit.remote (PennyLane-qiskit-0.32.0)
```

Existing GitHub issues

- [x] I have searched existing GitHub issues to make sure the issue does not already exist.

c-3543 added bug on Oct 7, 2023

isaacdevlugt on Oct 11, 2023 Contributor

Hey @c-3543! Thanks for making the bug report. Just for the paper trail, this came up as an issue in the PL-qiskit plugin because `QISKIT_OPERATION_MAP_SELF_ADJOINT` doesn't have a `'C(RZ)'` key but does have a `'CRZ'` one.

https://github.com/PennyLaneAI/pennylane-qiskit/blob/d6964c723278272196c9fdea57bd2da4db9344e9/pennylane_qiskit/qiskit_device.py#L36

albi3ro mentioned this on Oct 19, 2023

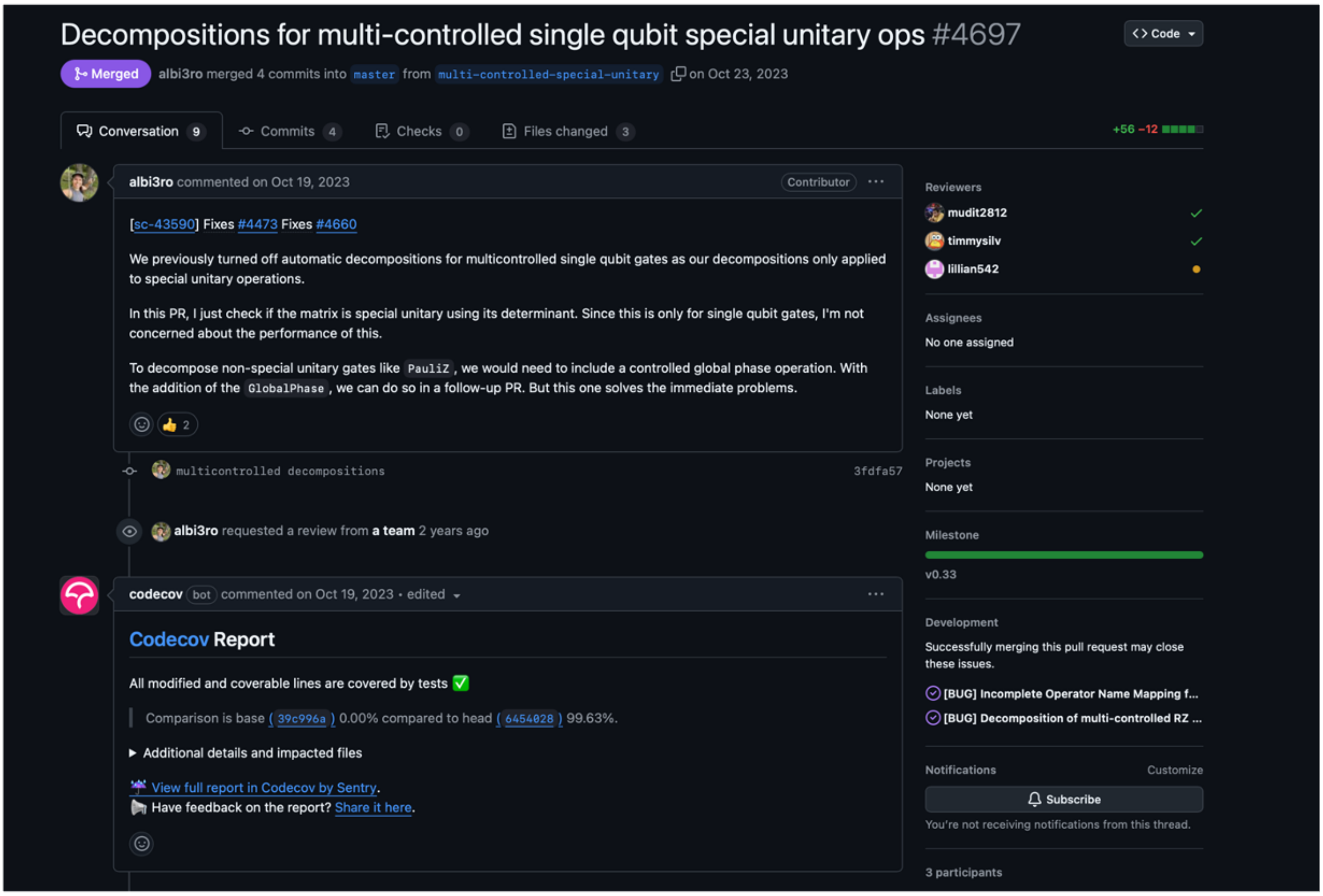
Decompositions for multi-controlled single qubit special unitary ops #4697

albi3ro closed this as completed in #4697 on Oct 23, 2023

albi3ro added a commit that references this issue on Oct 23, 2023

`Decompositions for multi-controlled single qubit special unitary ops` ... Verified a7114a9

Alex-Preciado changed the title ~~[BUG]~~ [BUG] Incomplete Operator Name Mapping for Controlled RX Gate on Nov 12, 2023


Decompositions for multi-controlled single qubit special unitary ops #4697
Code
Merged
albi3ro merged 4 commits into master from multi-controlled-special-unitary on Oct 23, 2023
Conversation 9
Commits 4
Checks 0
Files changed 3
+56 −12
albi3ro commented on Oct 19, 2023
Contributor
[sc-43590] Fixes #4473 Fixes #4660
We previously turned off automatic decompositions for multicontrolled single qubit gates as our decompositions only applied to special unitary operations.
In this PR, I just check if the matrix is special unitary using its determinant. Since this is only for single qubit gates, I'm not concerned about the performance of this.
To decompose non-special unitary gates like PauliZ, we would need to include a controlled global phase operation. With the addition of the GlobalPhase, we can do so in a follow-up PR. But this one solves the immediate problems.
2
multicontrolled decompositions
3fdfa57
albi3ro requested a review from a team 2 years ago
codecov bot commented on Oct 19, 2023 • edited
Codecov Report
All modified and coverable lines are covered by tests ✅
Comparison is base (39c996a) 0.00% compared to head (6454028) 99.63%.
Additional details and impacted files
View full report in Codecov by Sentry.
Have feedback on the report? Share it here.
Reviewers
mudit2812
timmysilv
lillian542
Assignees
No one assigned
Labels
None yet
Projects
None yet
Milestone
v0.33
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Successfully merging this pull request may close these issues.
[BUG] Incomplete Operator Name Mapping f...
[BUG] Decomposition of multi-controlled RZ ...
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